

Candidate Number _____

Fraction _____

2020 Republic of China Biology Olympiad National Team Selection Camp Practical Test Questions

Second Examination **Room**

Bioinformatics Practice

project	quantity
The laptop (with MEGA-X installed) has	1 unit
a COVID-2019 serial number file on its desktop	1
and a folder containing student exam numbers.	1

Please note:

1. Please check if the exam number on the exam paper matches your exam number.
2. This exam paper (including the cover and the question paper) has a total of 4 pages and must be returned in its entirety when submitting the exam.
3. The total time allotted for answering questions in this exam is 90 minutes.
4. Please answer the questions on this paper. You may write your answers on the back of the paper, but please indicate the question number.

The COVID-19 pandemic, which occurred globally from 2019 to 2020, has resulted in over 3 million infections worldwide. The COVID-2019.fas archive contains 17 sequencing results from samples obtained from patients and 2 bat viral sequence fragments. The following analysis was performed using the MEGA-X program. 1.

The results presented after aligning the 19 viral sequence data using Clustal W.

What problem do you think caused this phenomenon? (10 points)

2. After removing two bat virus sequences, what are the nucleotide chain lengths of the remaining 17 COVID-19 sequences obtained from patient samples? What are the number of conserved nucleotide sites? What are the number of variable nucleotide sites? What are the number of parsimony information nucleotide sites? What are the number of singleton nucleotide sites? (Note: COVID-2019 sequences need to be saved as MEGA and FAST files separately. Please save them to the folder with your exam number on your desktop, named "exam".)

"Exam Number - COVID-2019-1.MEG and Exam Number - COVID-2019-1.FAS" (2 points each, 10 points total)

project	Answer
Nucleotide chain length:	
Conserved(C):	
Mutation (Variable)(V):	
Parsimony information(Pi):	
Singleton(S):	

3. From the 17 obtained COVID-2019 sequences, locate the region from "ATGGAGAGC" to "AAACCTTT". Remove other regions. This fragment is the gene fragment of the ORF1a polyprotein of the SARS-CoV-2 virus. What is the nucleotide chain length of this fragment? What are the number of conserved nucleotide sites? What are the number of variable nucleotide sites? What are the number of parsimony information nucleotide sites? What are the number of singleton nucleotide sites? (Hint: The ORF1a-2019 sequence needs to be saved as MEGA and FAST files separately. Please save it to the folder with your exam number on your desktop, named "exam".)

Exam number - ORF1a-2019-1.MEG and exam number - ORF1a-2019-1.FAS. (2 points each, 10 points in

project	total) Answer
Nucleotide chain length:	
Conserved(C):	
Variable(V):	
Parsimony information(Pi):	
Singleton(S):	

4. After translation, what is the length of the amino acid chain of the ORF1a polyprotein gene fragment of the novel coronavirus?

How many conserved amino acid sites are there? How many variable amino acid sites are there?

Parsimony information: What are the amino acid sites? Singleton: What are the amino acid sites? (2 points each, 10 points total)

project	Answer
Amino acid chain length:	
Conserved(C):	
Variable(V):	
Parsimony information(Pi):	
Singleton(S):	

5. Gene fragments of 17 obtained COVID-2019 sequences and ORF1a polyprotein.

What are the average percentages of A, G, C, and T in the sequence? (2 points each, 16 points in total)

	A	G	C	T
COVID-2019				
ORF1a				

6. Seventeen COVID-19 sequences and ORF1a polyprotein gene fragments were obtained.

Using DNA sequences and based on the Bayesian Information Criterion (BIC) scores, select the top three most suitable nucleotide substitution models. Are there any differences between them? (2 points each, 12 points total)

	Top 1 model, top 2 models, top 3 models	
COVID-2019		
ORF1a		

Is there a difference? (6 points)

7. Using the gene fragment of the ORF1a polyprotein, and based on the amino acid sequence (Amino acid) and the BIC (Bayesian Information Criterion) score, select the top three nucleotide substitution models that best fit the criteria. (2 points each, 6 points in total)

	Top 1 model, top 2 models, top 3 models		
ORF1a			

8. Using the ORF1a polyprotein gene fragment, construct dendrograms for both DNA (nucleotide) and amino acid sequences, based on the obtained optimal nucleotide substitution models and amino acid substitution models, using the maximum likelihood method. Repeat the analysis 50 times using the bootstrap method to construct dendrograms for both DNA (nucleotide) and amino acid sequences. Please save the DNA dendrogram as a PDF file and list the DNA sequence clusters in parentheses. (5 points) Please save the amino acid dendrogram as a PDF file and list the amino acid sequence clusters in parentheses. (5 points) Please draw inferences based on the results of the two dendrograms. (10 points)